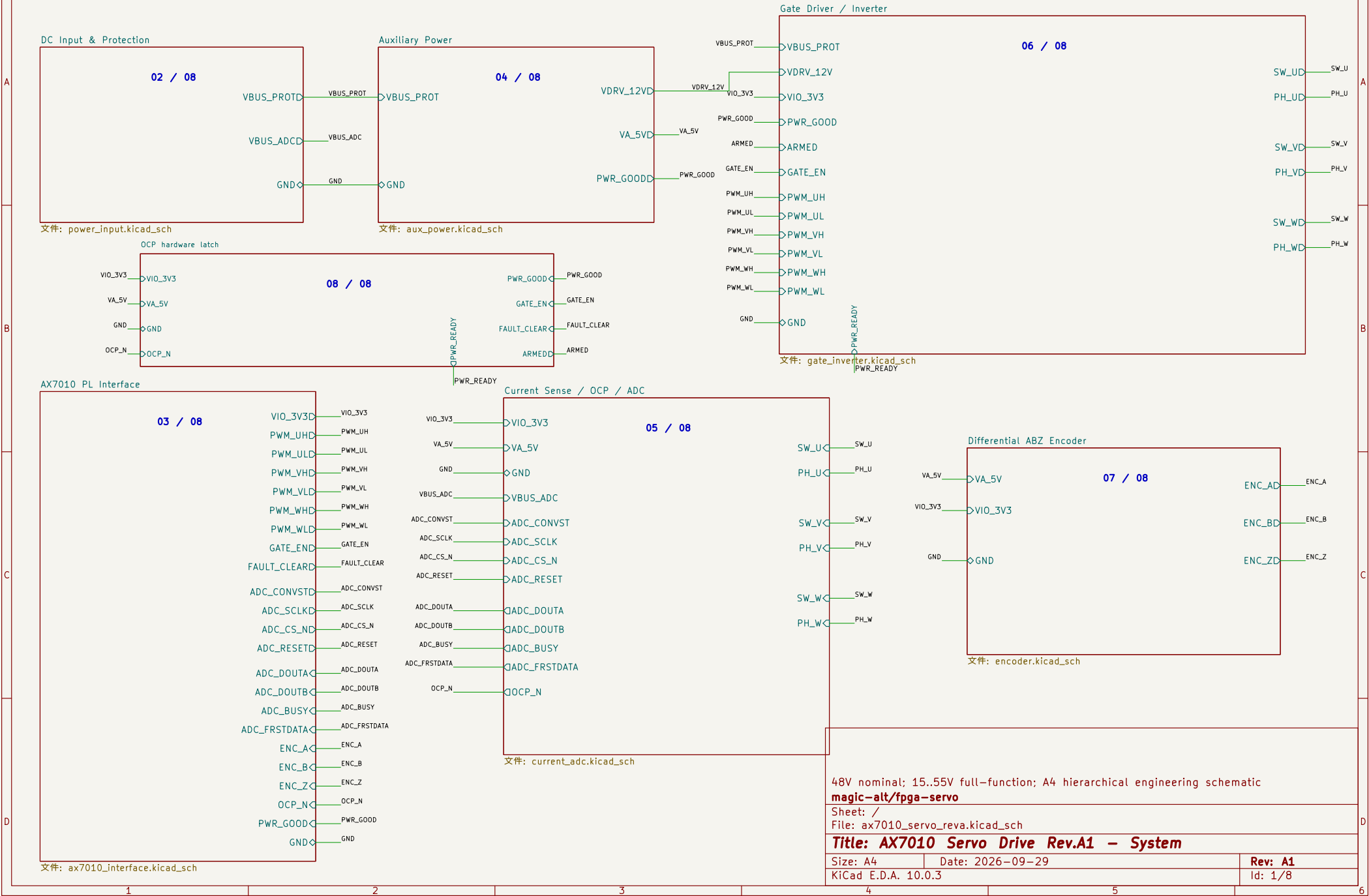
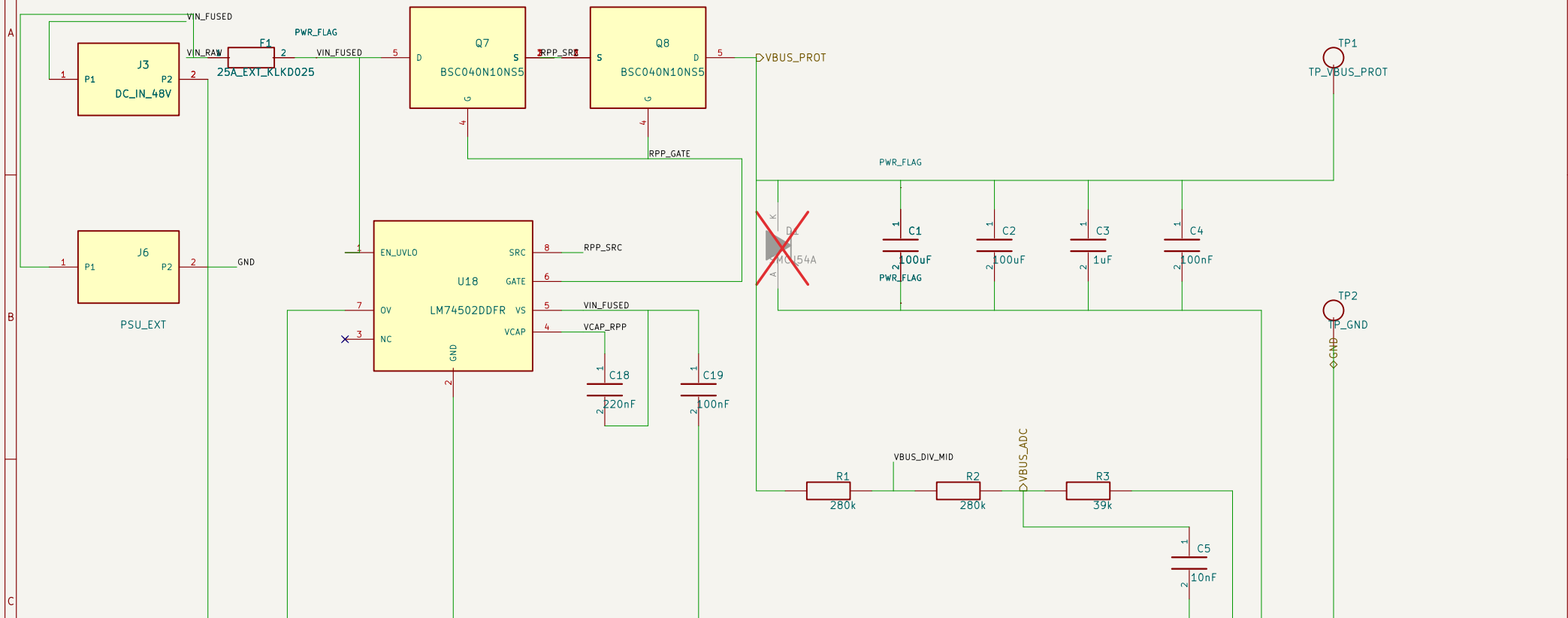


SYSTEM ARCHITECTURE – POWER FLOW / SAFETY FLOW / REAL-TIME I/O



SOURCE + -> EXTERNAL F1 -> J3.1 -> LM74502 RPP -> DC-LINK



Q7/Q8 are back-to-back N-MOSFETs driven by LM74502 for low-loss reverse-polarity protection. The controller has no reverse-current blocking when enabled, so regeneration can return to a receptive source. LM74502 OV is tied low in Rev.A1; positive surge is NOT claimed as closed. The 48V source/cable transient must be measured and kept inside the controller/downstream qualification envelope.
F1: EXTERNAL KLKD025.T + LPSM0001Z, source + to J3.1. No PCB fuse. Verify fault-current, inrush and thermal coordination. D1 stays DNP pending surge qualification.

VBUS_ADC divider = $280k + 280k \text{ over } 39k$, plus 10nF at the ADC node. Keep the high-side resistors split for voltage stress and route the sense return away from commutation current.
Local wiring: power, gate, divider/filter and ground. See docs/schematic_full_review_2026-10-09.md.

48V-class bus; 15..55V design window; source transient qualification required
magic-alt/fpga-servo

Sheet: /DC Input & Protection/
File: power_input.kicad_sch

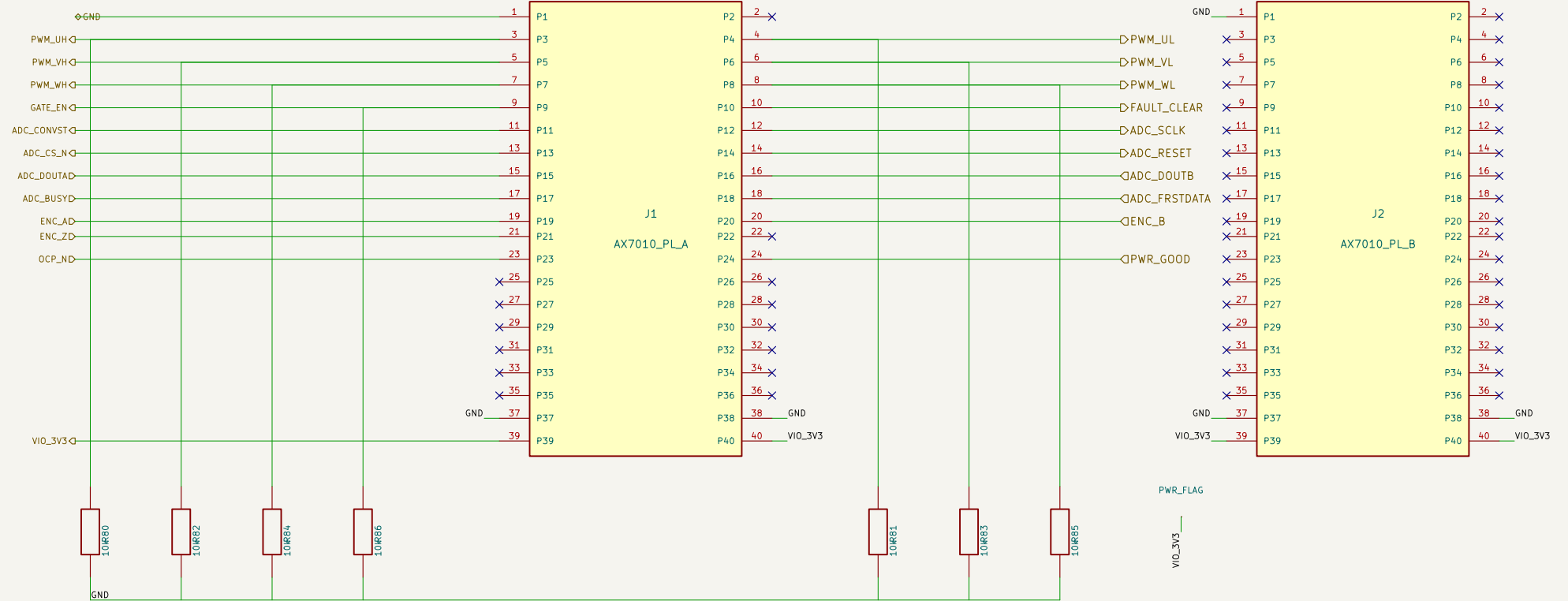
Title: Rev.A1 - DC Input & Protection

Size: A4	Date: 2026-09-29	Rev: A1
KiCad E.D.A. 10.0.3		Id: 2/8

ALINX AX7010 (2022): active J10 interface + physical reset-state pull-down branches

J1 = AX7010 J10

J2 = unused expansion (DNP)



R80..R86: each PWM and GATE_EN line has its own 10k pull-down.

J10 pin 2 (+5 V) remains NC: signal, power and ground pin mapping unchanged.

Two 2x20 PL expansion headers; recent AX7010 docs call these J10/J11
magic-alt/fpga-servo

Sheet: /AX7010 PL Interface/
File: ax7010_interface.kicad_sch

Title: Rev.A1 – AX7010 PL Interface

Size: A4

Date: 2026-09-29

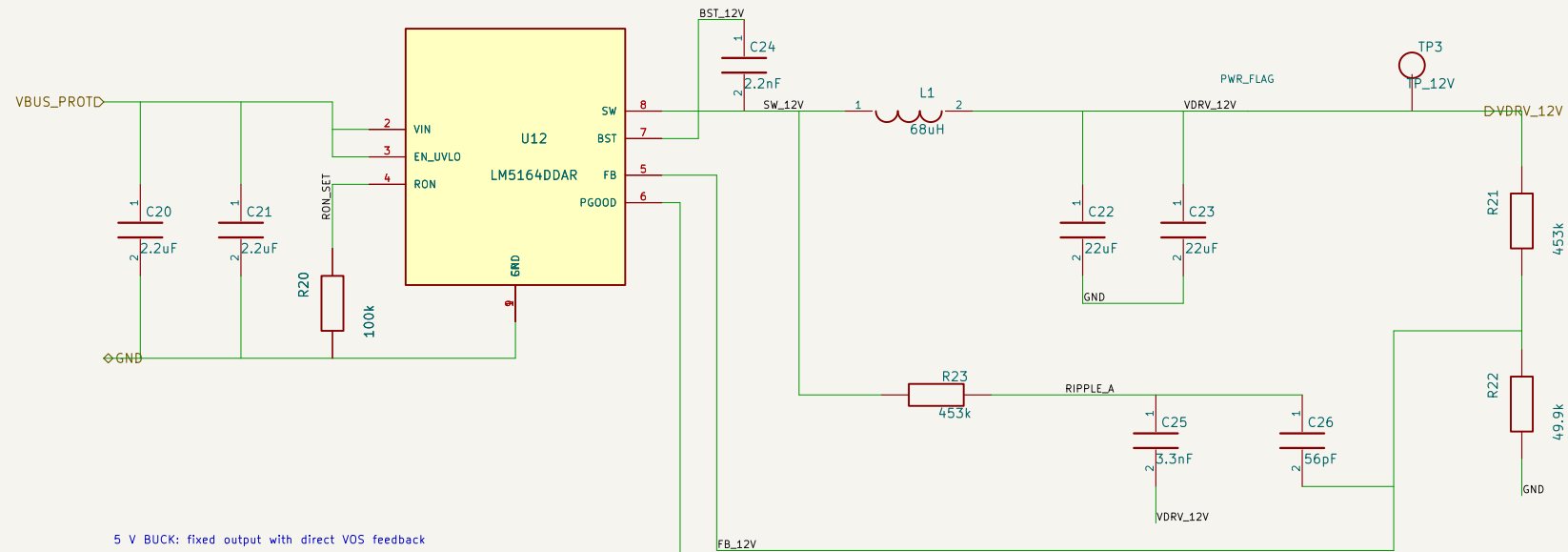
Rev: A1

KiCad E.D.A. 10.0.3

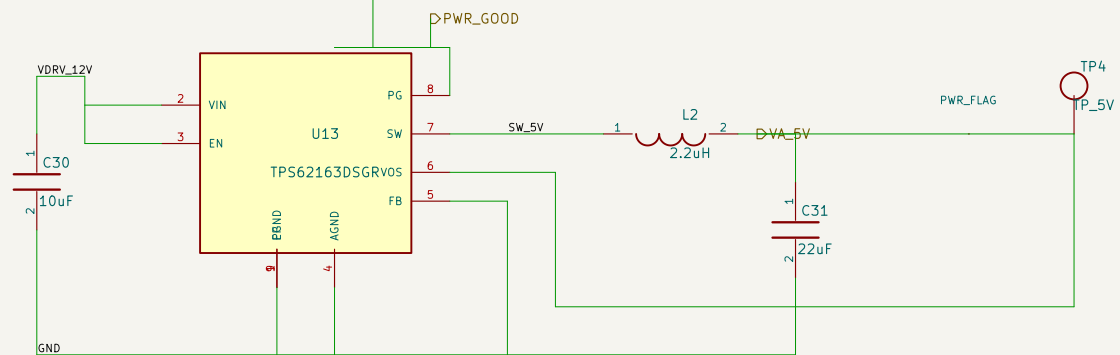
Id: 3/8

VBUS_PROT -> LM5164 12V / 1A -> TPS62163 5V / 1A. PGOOD outputs are wired-AND.

12 V BUCK: input / bootstrap / output LC / feedback + ripple injection



5 V BUCK: fixed output with direct VOS feedback



LM5164 values follow TI Figure 7-1: 15..100V In, 12V/1A, 300kHz, 68uH, 2x2.2uF CIN, 2x22uF COUT, 453k/49.9k feedback.

TPS62163 is the fixed 5V member of TPS6216x. Use 2.2uH + 22uF; connect FB to AGND and VOS to the 5V output.

LM5164 PGOOD and TPS62163 PG share PWR_GOOD; pull up at 3.3V safety logic.

12V input is for limited bring-up; full-function minimum is 15V.

Hierarchy ports attach to converter nets; DNP parts remain NoConn.

LM5164 48V->12V + TPS62163 fixed 5V

magic-alt/fpga-servo

Sheet: /Auxiliary Power/

File: aux_power.kicad_sch

Title: Rev.A1 - Auxiliary Power

Size: A4

Date: 2026-09-29

KiCad E.D.A. 10.0.3

Rev: A1

Id: 4/8

